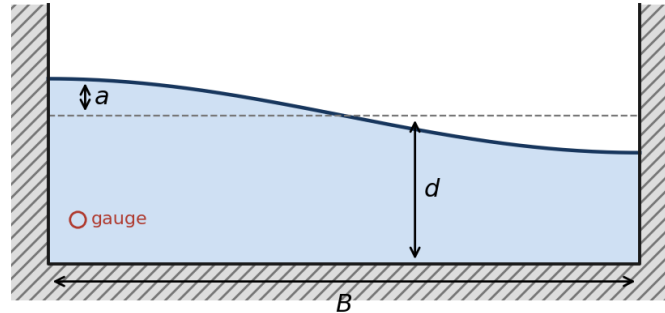


Tutorial: dimensional analysis and a sloshing tank

A design chart is wanted for the natural sloshing period T of rectangular tanks of length B holding water to depth d . An analyst plans to compute T by creating a set of models, with five values each of B , d and the initial tilt a of the surface (below). Viscosity and surface tension are negligible.



1. Dimensional analysis to guide model building

(a) How many models does the analyst plan?

(b) T depends on B , d , a , g and the density ρ . Find the Π groups.

(c) For small tilts the period does not depend on a . Show that T can then be written as

$$T = K\sqrt{d/g}, \text{ with } K = \phi(B/d),$$

and say how many models the design chart needs.

(d) In a long, shallow tank the surface disturbance travels at $\sqrt{gd}$ and goes the length of the tank and back in one period. Find K .

(e) Deep water is barely disturbed by a wave, so in a deep tank T does not depend on d . Use dimensional analysis to find an expression for T in a deep tank, and write it in terms of K from (c).

(f) Linear wave theory gives

$$T = 2\pi/\sqrt{gk \tanh(kd)}, \text{ with } k = \pi/B.$$

Show that

$$K_{\text{theory}} = 2\pi/\sqrt{kd \tanh(kd)}$$

depends on B/d only, and that it reduces to (d) and (e).

2. In the session (exercise DA-3)

A model gives a period T in seconds for one particular tank. Dividing by $\sqrt{d/g}$ removes the tank's size and leaves

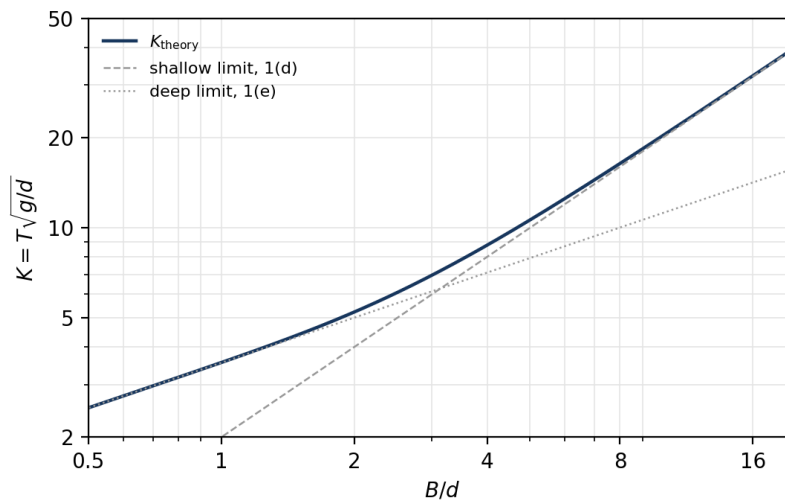
$$K_{\text{app}} = T\sqrt{g/d}$$

a pure number that can be compared directly with any other tank's, and with K_{theory} . By 1(c), one model per value of B/d is enough to draw the analyst's design chart: the five models below stand in for all 125.

Place a Depth gauge at $x = 0.5$ m, low in the water. For each model set Tank length and Water depth in Controls $\rightarrow$ Geometry, with Initial tilt $a/d = 0.2$. Expand the gauge ($\checkmark$) and read the times t_1 and t_4 of the first and fourth crests after release; $T = (t_4 - t_1)/3$.

Model	B (m)	d (m)	B/d	t_1 (s)	t_4 (s)	T (s)	K_{app}
A	1	1	1				
B	2	1	2				
C	4	1	4				
D	8	1	8				
E	8	0.5	16				

Plot K_{app} against B/d on the chart, which shows K_{theory} and the two limits from 1(d) and 1(e).



(a) Do your five points lie on one curve? How well does K_{theory} describe it?

(b) Without building a model, use the chart to find the period of a tank 20 m long holding 2.5 m of water.

Answers

1(a) 125

1(b) $T\sqrt{g/d}$, B/d , a/d

1(c) 5 models

1(d) $K = 2B/d$

1(e) $T = C\sqrt{B/g}$, so $K = C\sqrt{B/d}$

1(f) shallow: $K \rightarrow 2B/d$; deep: $K \rightarrow 2\sqrt{\pi}\sqrt{B/d}$, so $C = 2\sqrt{\pi}$

2 $K_{\text{app}} \approx 3.8, 5.4, 8.9, 17, 33$ for models A–E

2(a) yes, within a few per cent of K_{theory}

2(b) $K_{\text{theory}} = 16.4$, so $T = 8.3$ s